

Python-Social Media Engagement Data Analysis

Objectives:

The analyze the Social Media Engagement dataset and identify meaningful patterns in user engagement, content performance, user behavior, posting time, device usage, and sentiment.

The project aims to clean and prepare the data, perform statistical and exploratory analysis, visualize important patterns, and generate useful insights from social media activities such as likes, comments, shares, impressions, watch time, followers, and engagement rate.

Task 1

Dataset Provided: social_media_engagement_5000.csv

Data Import & Setup

- Import CSV using Pandas
- Check/convert data types
- Convert date columns to datetime

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The notebook has two code cells. The first cell imports the necessary libraries: pandas, numpy, plotly.express, plotly.graph_objects, matplotlib.pyplot, and seaborn. The second cell loads the dataset from a GitHub repository using pd.read_csv. Below the code, the first two rows of the dataset are displayed as a table.

	user_id	age	gender	country	post_id	post_type	post_category	likes	comments	shares	watch_time_sec	impression_count	posted_at
0	25795	43.0	Female	Brazil	496713	image	fitness	7011.0	354.0	1157.0	5726	44650	17-12-2022
1	10860	33.0	Male	Brazil	157326	reel	food	11750.0	2606.0	1807.0	5947	80216	02-06-2025

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code is as follows:

```
'device_type', 'sentiment', 'hashtags', 'engagement_rate'],
dtype='object')

[76]
✓ 0s
1 #Convert date columns to datetime
2 df.loc[:, "posted_at"] = pd.to_datetime(
3     df["posted_at"], errors="coerce"
4 )

/tmp/ipykernel_2955/1492980354.py:2: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `day
df.loc[:, "posted_at"] = pd.to_datetime(

[134]
✓ 0s
1 df = df.copy()
2
3 # Fill missing age values with median
4 df["age"] = df["age"].fillna(df["age"].median())
5
6 # Convert age to integer
7 df["age"] = df["age"].astype(int)

[138]
✓ 0s
1 df.info()
```

The bottom of the notebook shows the "Variables" tab with the following information:

```
18 engagement_rate  5000 non-null  float64
dtypes: bool(1), datetime64[ns](1), float64(4), int64(6), object(7)
memory usage: 708.1+ KB
```

Task 2 — Data Cleaning

• Cleaning Missing Data

■ Detect missing values (isnull(), isna())

■ Handle using: dropna(), fillna(), median/mode, forward/backward-fill

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code is as follows:

```
18 engagement_rate  5000 non-null  float64
dtypes: bool(1), datetime64[ns](1), float64(4), int64(6), object(7)
memory usage: 708.1+ KB

[11]
✓ 0s
1 df.isnull().sum()
```

The output of the code is a series of counts for missing values in each column:

Column	Count
user_id	0
age	0
gender	150
country	0
post_id	0
post_type	0
post_category	0
likes	150
comments	150

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The notebook is open to a cell containing the following Python code:

```
hashtags    0.0
engagement_rate 0.0
dtype: float64
```

```
[13] 1 #handle Missing Values
      2 categorical_columns = df.select_dtypes(include="object").columns
      3
      4 for col in categorical_columns:
      5     df[col] = df[col].fillna(df[col].mode()[0])
```

```
[14] 1 numeric_columns = df.select_dtypes(include=np.number).columns
      2
      3 for col in numeric_columns:
      4     df[col] = df[col].fillna(df[col].median())
```

```
[15] 1 df.isnull().sum()
```

The output of the code shows the data types of the columns and the result of the missing value handling. The notebook interface includes a left sidebar with icons for file explorer, search, and other tools. The bottom status bar shows the time as 7:55 AM and the Python version as Python 3.

The screenshot shows the same Google Colab notebook, now displaying the final dataset structure. The output of the code is a list of columns and their corresponding values, all set to 0:

```
gender      0
country     0
post_id     0
post_type   0
post_category 0
likes       0
comments    0
shares      0
watch_time_sec 0
impression_count 0
posted_at   0
follower_count 0
is_verified 0
device_type 0
```

The notebook interface is consistent with the previous screenshot, showing the same file explorer and status bar.

- **Duplicate Handling- Identify & remove duplicates**

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with "Commands", "+ Code", "+ Text", and "Run all", and a status bar showing "7:55 AM" and "Python 3".

The notebook content is as follows:

- A table of statistics:

std	14.687151
min	13.000000
25%	26.000000
50%	38.000000
75%	51.000000
max	64.000000
- Code cell [19] with the following code:

```
1 #duplicate handling
2 df.duplicated().sum()
np.int64(0)
```
- Code cell [] with the following code:

```
1
```
- A section titled "Data Formatting".

The bottom of the notebook shows a "Variables" tab and a "Terminal" tab.

• Data Formatting

■ Fix incorrect data types

The screenshot shows the same Google Colab notebook, now with more code under the "Data Formatting" section.

- Code cell [20] with the following code:

```
1 #fix incorrect data types
2 df["likes"] = pd.to_numeric(df["likes"], errors="coerce")
3 df["comments"] = pd.to_numeric(df["comments"], errors="coerce")
4 df["shares"] = pd.to_numeric(df["shares"], errors="coerce")
5
6 df["engagement_rate"] = pd.to_numeric(
7     df["engagement_rate"], errors="coerce"
8 )
```
- Code cell [21] with the following code:

```
1 #Standardize Gender
2 df["gender"].value_counts()
```
- The output of cell [21] shows a table:

gender	count
Male	1849
Female	1504

The bottom of the notebook shows the "Variables" tab and "Terminal" tab, with the time updated to "8:03 AM".

■ Standardize categories (e.g., gender labels)

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell [22] contains the following Python code:

```
1 df["gender"] = df["gender"].str.strip().str.lower()
```

The code cell [23] contains the following Python code:

```
1 df["gender"].value_counts()
```

The output of the code cell [23] is a value_counts table for the 'gender' column:

gender	count
male	1849
other	1581
female	1570

The notebook interface includes a left sidebar with navigation icons, a top bar with the file name and a "Share" button, and a bottom bar with a terminal and system status.

■ Correct unrealistic values in likes, comments, shares

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell [25] contains the following Python code:

```
1 for col in ["likes", "comments", "shares"]:
2     df[col] = df[col].fillna(df[col].median())
```

The code cell [26] contains the following Python code:

```
1 df.loc[(df["age"] < 13) | (df["age"] > 100), "age"] = np.nan
```

The code cell [27] contains the following Python code:

```
1 df[["likes", "comments", "shares"]].describe()
```

The output of the code cell [27] is a describe table for the 'likes', 'comments', and 'shares' columns:

	likes	comments	shares
count	5000.000000	5000.000000	5000.000000
mean	10106.997400	1502.039800	1002.9106
std	5702.293017	856.393312	570.8552
min	10.000000	0.000000	0.000000
25%	5235.000000	792.000000	511.000000
50%	10105.500000	1497.000000	1012.000000
75%	14959.000000	2235.250000	1483.000000
max	19998.000000	2999.000000	1999.000000

The notebook interface includes a left sidebar with navigation icons, a top bar with the file name and a "Share" button, and a bottom bar with a terminal and system status.

● Feature Cleaning

■ Extract hashtag count

■ Clean sentiment labels

Python-Social Media Engagement Data Analysis.ipynb

```

75% 14959.000000 2235.250000 1483.0000
max 19998.000000 2999.000000 1999.0000

[28] ✓
1 #Feature Cleaning
2 #Extract hashtag count
3 df["hashtag_count"] = df["hashtags"].fillna("").str.count("#")
4 # Clean sentiment labels
5 df["sentiment"] = df["sentiment"].str.strip().str.lower()

[29] ✓ 0s
1 df[["hashtags", "hashtag_count"]].head()

```

	hashtags	hashtag_count
0	#foodie #travel #love	3
1	#fitness	1
2	#foodie	1
3	#music #foodie #fun	3

Task 3 — Data Exploration using Pandas

- View dataset structure using `head()`, `tail()`, `shape`, and `columns`.

Python-Social Media Engagement Data Analysis.ipynb

```

[30] ✓ 0s
1 #View dataset structure using head(), tail(), shape, and columns.
2 df.head()
3

```

	user_id	age	gender	country	post_id	post_type	post_category	likes	comments	shares	watch_time_sec	impression_count	posted_at	...
0	25795	43.0	female	Brazil	496713	image	fitness	7011.0	354.0	1157.0	5726	44650	2022-12-17	
1	10860	33.0	male	Brazil	157326	reel	food	11750.0	2606.0	1807.0	5947	80216	2023-06-02	
2	86820	32.0	female	UK	109864	text	food	4862.0	344.0	955.0	6946	44858	2023-05-07	

Python-Social Media Engagement Data Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[31] ✓ 0s
1 df.tail()
```

	user_id	age	gender	country	post_id	post_type	post_category	likes	comments	shares	watch_time_sec	impression_count	posted_at
4995	59500	44.0	male	Australia	441541	video	education	16210.0	2013.0	1837.0	6190	42977	2022-06-25
4996	22100	38.0	other	UAE	677076	reel	education	16924.0	2734.0	1583.0	7764	34196	2022-11-18
4997	67021	63.0	female	USA	273595	text	travel	13487.0	1497.0	167.0	7466	23680	2023-04-06
4998	29800	13.0	female	Germany	785644	video	fitness	16894.0	1289.0	1713.0	4991	89013	2022-05-16
4999	73400	54.0	other	Japan	712252	text	travel	14830.0	503.0	1798.0	3743	14234	2023-03-04

Variables Terminal 7:55 AM Python 3

Type here to search

Python-Social Media Engagement Data Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[32] ✓ 0s
1 df.shape
(5000, 20)

[33] ✓ 0s
1 df.columns
Index(['user_id', 'age', 'gender', 'country', 'post_id', 'post_type',
      'post_category', 'likes', 'comments', 'shares', 'watch_time_sec',
      'impression_count', 'posted_at', 'follower_count', 'is_verified',
      'device_type', 'sentiment', 'hashtags', 'engagement_rate',
      'hashtag_count'],
      dtype='object')

[34] ✓ 0s
1 #Check data types and info with info() and dtypes.
2 df.dtypes
```

Variables Terminal 7:55 AM Python 3

Type here to search

- Check data types and info with info() and dtypes.

Python-Social Media Engagement Data Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

(5000, 20)

[101] 1 df.columns

Index(['user_id', 'age', 'gender', 'country', 'post_id', 'post_type', 'post_category', 'likes', 'comments', 'shares', 'watch_time_sec', 'impression_count', 'posted_at', 'follower_count', 'is_verified', 'device_type', 'sentiment', 'hashtags', 'engagement_rate', 'hashtag_count'], dtype='object')

[139] 1 #Check data types and info with info() and dtypes.
2 df.dtypes

user_id	int64	
age	int64	
gender	object	

Variables Terminal

8:19 AM Python 3

Python-Social Media Engagement Data Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

[157] 1 df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   user_id                5000 non-null   int64
1   age                   5000 non-null   float64
2   gender                 5000 non-null   object
3   country                5000 non-null   object
4   post_id                5000 non-null   int64
5   post_type              5000 non-null   object
6   post_category          5000 non-null   object
7   likes                  5000 non-null   float64
8   comments                5000 non-null   float64
9   shares                 5000 non-null   float64
10  watch_time_sec         5000 non-null   int64
11  impression_count        5000 non-null   int64
12  posted_at               5000 non-null   object
13  follower_count          5000 non-null   int64
14  is_verified             5000 non-null   bool
15  device_type             5000 non-null   object
16  sentiment               5000 non-null   object
```

Google Drive

Uploads can't complete

Your last successful upload was 37 days ago. Syncing is currently paused. You may want to resume syncing to ensure your files are uploading successfully.

OK Resume syncing

Variables Terminal

8:26 AM Python 3

- Generate summary statistics using describe().

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell contains the command `df.describe().T`, which has been executed. The output is a summary statistics table for the dataset.

	count	mean	std	min	25%	50%	75%	max
user_id	5000.0	54561.890800	26090.370121	10055.000000	32309.500000	54374.500000	77180.500000	99963.000000
age	5000.0	38.440400	14.687151	13.000000	26.000000	38.000000	51.000000	64.000000
post_id	5000.0	548042.909000	260646.957267	100068.000000	322543.500000	548077.500000	771574.500000	999455.000000
likes	5000.0	10106.997400	5702.293017	10.000000	5235.000000	10105.500000	14959.000000	19998.000000
comments	5000.0	1502.039800	856.393312	0.000000	792.000000	1497.000000	2235.250000	2999.000000
shares	5000.0	1002.910600	570.855200	0.000000	511.000000	1012.000000	1483.000000	1999.000000
watch_time_sec	5000.0	4014.503200	2308.096459	0.000000	2017.750000	4034.500000	6020.250000	7998.000000
impression_count	5000.0	50013.732800	28844.939104	105.000000	24988.250000	49934.500000	74662.250000	99995.000000
follower_count	5000.0	393698.224800	230927.884535	87.000000	194480.000000	388982.000000	589744.250000	799533.000000
engagement_rate	5000.0	0.964356	5.318029	0.006363	0.145781	0.253896	0.504794	191.504348
hashtag_count	5000.0	1.998600	0.812853	1.000000	1.000000	2.000000	3.000000	3.000000

The screenshot shows the same Google Colab notebook, but now displaying the "Interpretation" section. The code cell contains the command `df.describe().T`, which has been executed. The output is a summary statistics table for the dataset.

	count	mean	std	min	25%	50%	75%	max
engagement_rate	5000.0	0.964356	5.318029	0.006363	0.145781	0.253896	0.504794	191.504348
hashtag_count	5000.0	1.998600	0.812853	1.000000	1.000000	2.000000	3.000000	3.000000

Interpretation

The dataset contains 5,000 social media records covering posts from 2022 to 2023. The average user age is 38.44 years. Posts receive approximately 10,107 likes, 1,502 comments, and 1,003 shares on average. The average impression count is around 50,014, while the average follower count is approximately 393,698. Most numerical variables have mean and median values that are relatively close, indicating fairly balanced distributions. However, the engagement_rate column has a very high maximum value of 191.50 compared with its median of 0.254, suggesting the presence of potential outliers that should be investigated during data cleaning.

Analyze categorical distributions using `value_counts()`, `unique()`, and `nunique()`.

```

1 # Find categorical columns
2 categorical_columns = df.select_dtypes(include="object").columns

```

- Analyze categorical distributions using `value_counts()`, `unique()`, and `nunique()`.

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell displays the output of two value count operations. The first operation is for the 'gender' column, showing counts for 'male' (1849), 'other' (1581), and 'female' (1570). The second operation is for the 'country' column, showing counts for various countries including India (535), Canada (513), Brazil (504), UAE (503), USA (501), France (496), UK (493), and Australia (493).

```

--- gender ---
Value Counts:
gender
male      1849
other     1581
female    1570
Name: count, dtype: int64
Unique Values:
['female' 'male' 'other']
Number of Unique Values:
3

--- country ---
Value Counts:
country
India      535
Canada     513
Brazil     504
UAE        503
USA        501
France     496
UK         493
Australia  493

```

• Create a correlation matrix for numeric fields.

The screenshot shows the same Google Colab notebook, but now displaying a correlation matrix for numeric fields. The matrix is a 10x10 table with the following columns: user_id, age, post_id, likes, comments, shares, watch_time_sec, impression_count, follower_count, and engagement_rate. The diagonal elements are all 1.000000, indicating perfect self-correlation. The off-diagonal elements represent the correlation between different fields, with values ranging from -0.16847 to 0.0935.

	user_id	age	post_id	likes	comments	shares	watch_time_sec	impression_count	follower_count	engagement_rate
user_id	1.000000	-0.006688	0.020051	0.025811	-0.033395	0.013763	-0.016847	0.015326	0.010124	-0.0042
age	-0.006688	1.000000	-0.013153	-0.036322	-0.007284	0.013871	0.005542	0.013322	-0.024894	0.0080
post_id	0.020051	-0.013153	1.000000	0.014526	-0.010540	0.001846	0.018374	-0.007709	-0.002844	0.0101
likes	0.025811	-0.036322	0.014526	1.000000	-0.018421	0.004712	0.008710	0.007952	-0.022982	0.0935
comments	-0.033395	-0.007284	-0.010540	-0.018421	1.000000	0.006142	-0.016351	-0.009395	-0.011733	0.0000
shares	0.013763	0.013871	0.001846	0.004712	0.006142	1.000000	0.014658	-0.005204	-0.010783	0.0217
watch_time_sec	-0.016847	0.005542	0.018374	0.008710	-0.016351	0.014658	1.000000	-0.004335	0.002761	-0.0011
impression_count	0.015326	0.013322	-0.007709	0.007952	-0.009395	-0.005204	-0.004335	1.000000	-0.015513	-0.2322
follower_count	0.010124	-0.024894	-0.002844	-0.022982	-0.011733	-0.010783	0.002761	-0.015513	1.000000	0.0022
engagement_rate	-0.004282	0.008039	0.010139	0.093520	0.000051	0.021724	-0.001148	-0.232226	0.002292	1.0000
hashtag_count	-0.013692	0.007173	0.007344	-0.002190	-0.015230	0.013379	-0.023301	-0.002714	0.008802	0.0053

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The top toolbar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the toolbar, there are tabs for "Commands", "Code", "Text", and "Run all". The main content area displays a correlation matrix for various features, with "hashtag_count" as the first column. The matrix shows values ranging from -0.2322 to 0.0935. Below the matrix, there is a section titled "Interpretation" which states: "The correlation analysis shows that most numerical variables have very weak relationships. The strongest correlation is between impression_count and engagement_rate (-0.2322), indicating a weak negative relationship. likes and engagement_rate have a very weak positive correlation (0.0935). Overall, the dataset does not show strong linear relationships among the numerical variables." Below this, there is a section titled "Groupby" with a code cell containing the following Python code:

```
[108]
✓ 0s
1 #Average like by post type
2 df.groupby("post_type")["likes"].mean()
```

The bottom status bar shows the time as 8:26 AM and the environment as Python 3.

- Use `groupby()` to summarize metrics (e.g., avg likes by post type, impressions by country).

The screenshot shows the same Google Colab notebook, but now the "Groupby" section is expanded. The code cell from the previous screenshot is still visible. Below it, the output of the `df.groupby("post_type")["likes"].mean()` operation is displayed as a table:

post_type	likes
image	10104.865277
reel	10037.802416
text	10100.148193
video	10188.600000

The output also shows the data type as `dtype: float64`. The bottom status bar shows the time as 8:26 AM and the environment as Python 3.

Python-Social Media Engagement Data Analysis.ipynb

```
1 #sentiment performance
2 df.groupby("sentiment")[
3     ["likes", "comments", "shares", "engagement_rate"]
4 ].mean().sort_values(
5     "engagement_rate",
6     ascending=False
7 )
```

	likes	comments	shares	engagement_rate
negative	10208.096875	1516.094792	1007.307292	1.038486
neutral	9923.194499	1528.404060	996.565160	0.991170
positive	10180.062077	1480.650617	1005.086749	0.919744

Data Wrangling

Python-Social Media Engagement Data Analysis.ipynb

```
1 #average engagement rate by country
2 df.groupby("country")["engagement_rate"].mean()
3
```

country	engagement_rate
Australia	1.324339
Brazil	1.540704
Canada	0.916659
France	1.146402
Germany	0.759000
India	0.655005
Japan	0.769623
UAE	1.112352

Task 4 — Data Wrangling

- Use merge, concat, or join if combining DataFrames.

Python-Social Media Engagement Data Analysis.ipynb

```
1 # Use merge, concat, or join if combining DataFrames
2 df1 = df[["user_id", "age", "gender"]]
3 df2 = df[["user_id", "country", "device_type"]]
4
5 merged_df = pd.merge(df1, df2, on="user_id")
6
7 merged_df.head()
```

	user_id	age	gender	country	device_type
0	25795	43.0	female	Brazil	mobile
1	10860	33.0	male	Brazil	mobile
2	86820	32.0	female	UK	tablet
3	64886	51.0	other	France	mobile
4	16265	34.0	other	UK	mobile

Python-Social Media Engagement Data Analysis.ipynb

Interpretation

merge() combines two DataFrames using the common column user_id.

```
1
2 df1 = df.iloc[:100]
3 df2 = df.iloc[100:200]
4
5 combined_df = pd.concat([df1, df2], ignore_index=True)
6
7 combined_df.head()
```

	user_id	age	gender	country	post_id	post_type	post_category	likes	comments	shares	watch_time_sec	impression_count	posted_at	fr
0	25795	43.0	female	Brazil	496713	image	fitness	7011.0	354.0	1157.0	5726	44650	2022-12-17	
1	10860	33.0	male	Brazil	157326	reel	food	11750.0	2606.0	1807.0	5947	80216	2023-06-02	
2	86820	32.0	female	UK	109864	text	food	4862.0	344.0	955.0	6946	44858	2023-05-07	

Python-Social Media Engagement Data Analysis.ipynb

```
1 #Perform groupby summaries by post_type, country, and sentiment.
2
3 #groupby-post Type
4 post_type_summary = df.groupby("post_type").agg(
5     avg_likes=("likes", "mean"),
6     avg_comments=("comments", "mean"),
7     avg_shares=("shares", "mean"),
8     avg_engagement=("engagement_rate", "mean")
9 )
10 post_type_summary
```

	avg_likes	avg_comments	avg_shares	avg_engagement
post_type				
image	10104.865277	1525.235766	1019.289495	0.895646
reel	10037.802416	1504.187062	982.042089	0.783047
text	10100.148193	1499.106024	1013.989558	1.064549
video	10188.600000	1479.160000	996.834286	1.122365

Python 3

Python-Social Media Engagement Data Analysis.ipynb

```
1 #groupby country
2 country_summary = df.groupby("country").agg(
3     avg_likes=("likes", "mean"),
4     avg_comments=("comments", "mean"),
5     avg_shares=("shares", "mean"),
6     avg_engagement=("engagement_rate", "mean")
7 )
8
9
10 country_summary
```

	avg_likes	avg_comments	avg_shares	avg_engagement
country				
Australia	10294.124746	1485.803245	1029.693712	1.324339
Brazil	9886.689484	1490.434524	1008.859127	1.540704
Canada	10063.066277	1546.582846	996.245614	0.916659
France	10372.876008	1478.695565	1051.812500	1.146402

Python 3

Python-Social Media Engagement Data Analysis.ipynb

```

1 #group by sentiment
2 sentiment_summary = df.groupby("sentiment").agg(
3     avg_likes=("likes", "mean"),
4     avg_comments=("comments", "mean"),
5     avg_shares=("shares", "mean"),
6     avg_engagement=("engagement_rate", "mean")
7 )
8
9 sentiment_summary

```

	avg_likes	avg_comments	avg_shares	avg_engagement
negative	10208.096875	1516.094792	1007.307292	1.038486
neutral	9923.194499	1528.404060	996.565160	0.991170
positive	10180.062077	1480.650617	1005.086749	0.919744

Task 5 — Statistical Analysis

Compute descriptive stats for “likes, comments, shares, watch_time_sec, engagement_rate, follower_count” columns:

- Mean, median, mode
- Standard deviation, variance
- Percentiles
- (Optional) Skewness and kurtosis

Python-Social Media Engagement Data Analysis.ipynb

```

11 "Kurtosis": df[stats_columns].kurtosis()
12 })
13
14 statistics_summary

```

	Mean	Median	Mode	Std Dev	Variance	25%	50%	75%	Skewness
likes	10106.997400	10105.500000	10105.500000	5702.293017	3.251615e+07	5235.000000	10105.500000	14959.000000	-0.006894
comments	1502.039800	1497.000000	1497.000000	856.393312	7.334095e+05	792.000000	1497.000000	2235.250000	0.003802
shares	1002.910600	1012.000000	1012.000000	570.855200	3.258757e+05	511.000000	1012.000000	1483.000000	-0.014654
watch_time_sec	4014.503200	4034.500000	916.000000	2308.096459	5.327309e+06	2017.750000	4034.500000	6020.250000	-0.018196
engagement_rate	0.964356	0.253896	0.006363	5.318029	2.828143e+01	0.145781	0.253896	0.504794	18.781271
follower_count	393698.224800	388982.000000	497502.000000	230927.884535	5.332769e+10	194480.000000	388982.000000	589744.250000	0.041118

Next steps: [New interactive sheet](#)

Python-Social Media Engagement Data Analysis.ipynb

Next steps: [New interactive sheet](#)

Statistical Interpretation

Likes: Mean and median are almost equal (~10,106), with skewness near 0, indicating a nearly symmetric distribution.

Comments: Average is ~1,502 and median is ~1,497. Skewness is close to 0, showing a balanced distribution.

Shares: Mean (1,003) and median (~1,012) are close, indicating little skewness.

Watch Time: Mean is ~4,015 seconds and median ~4,035 seconds. The distribution is fairly symmetric.

Engagement Rate: Mean (0.964) is much higher than median (0.254), with very high positive skewness (18.78) and kurtosis (482.99), indicating strong outliers and a highly right-skewed distribution.

Follower Count: Mean (393,698) and median (~388,982) are close, with skewness near 0, suggesting a fairly balanced distribution.

Likes, comments, shares, watch time, and follower count are relatively normally distributed, while engagement rate is highly skewed and contains significant outliers.

```
1 #Engagement Rate: Mean (0.964) is much higher than median (0.254) with very high positive skewness (18.78) and kurtosis(482.99)
```

Python-Social Media Engagement Data Analysis.ipynb

```
1 #Engagement Rate: Mean (0.964) is much higher than median (0.254), with very high positive skewness (18.78) and kurtosis(482.99),
2 # indicating strong outliers and a highly right-skewed distribution.
3
4 #I used the IQR (Interquartile Range) method to detect and remove outliers from the "engagement_rate" column.
5 #Values below the lower limit or above the upper limit were considered outliers and removed
6
7 Q1 = df["engagement_rate"].quantile(0.25)
8 Q3 = df["engagement_rate"].quantile(0.75)
9
10 IQR = Q3 - Q1
11
12 lower_limit = Q1 - 1.5 * IQR
13 upper_limit = Q3 + 1.5 * IQR
14
15 df = df[
16     (df["engagement_rate"] >= lower_limit) &
17     (df["engagement_rate"] <= upper_limit)
18 ]
19
20 print(df["engagement_rate"].describe())
21
```

Python-Social Media Engagement Data Analysis.ipynb

```

20 print(df["engagement_rate"].describe())
21
22
23

```

count	4422.000000
mean	0.289277
std	0.216163
min	0.006363
25%	0.133880
50%	0.226525
75%	0.373339
max	1.042766

Name: engagement_rate, dtype: float64

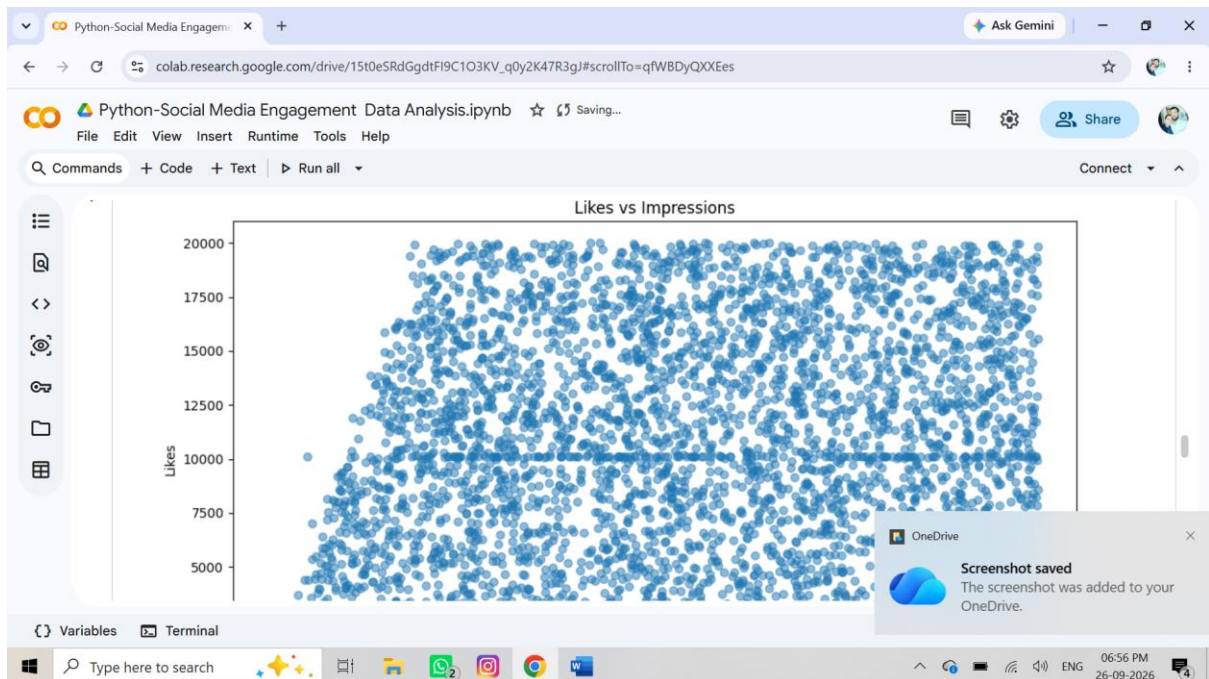
interpretation: After removing outliers, the mean engagement rate decreased from 0.964 to 0.289, and the mean and median are now much closer. This indicates that extreme values were successfully reduced, resulting in a more balanced distribution.

1

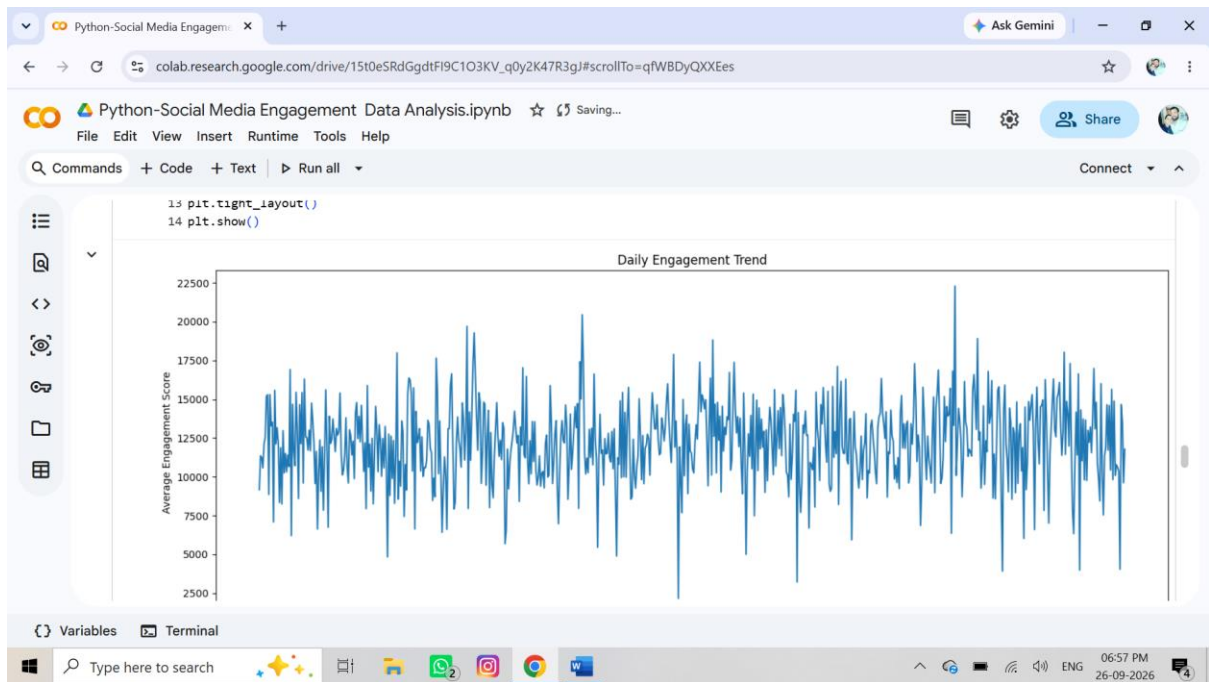
Task 6 — Data Visualization

• Matplotlib

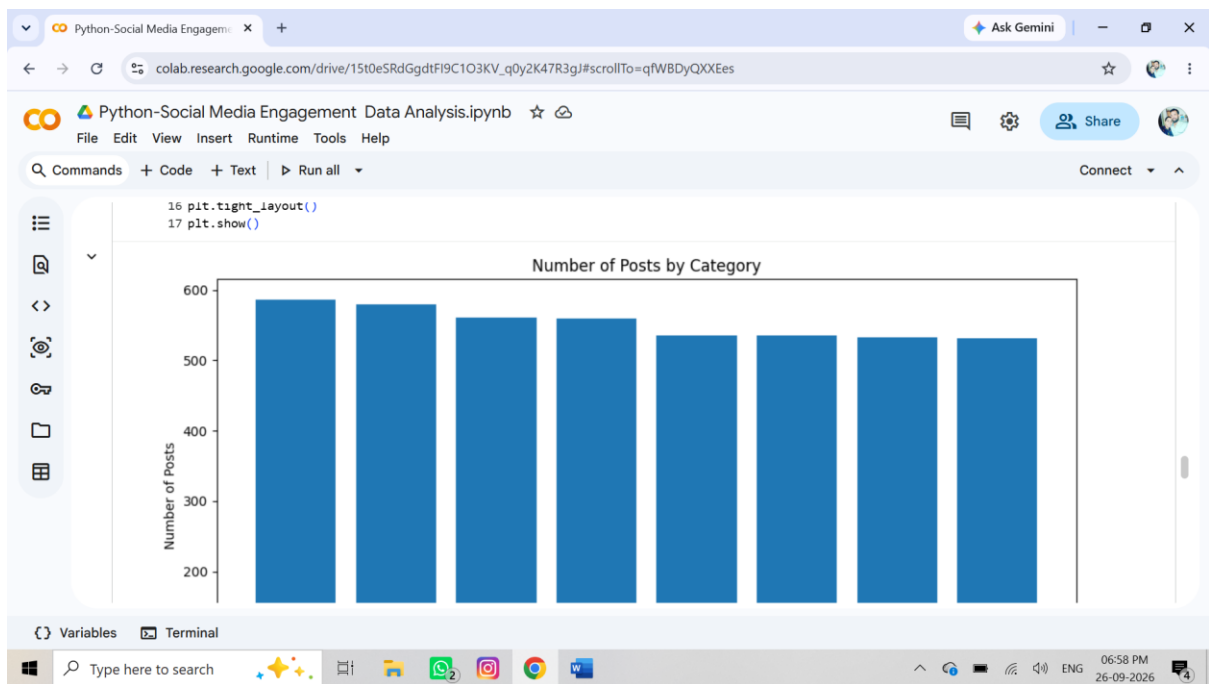
○ Scatter: likes vs impressions



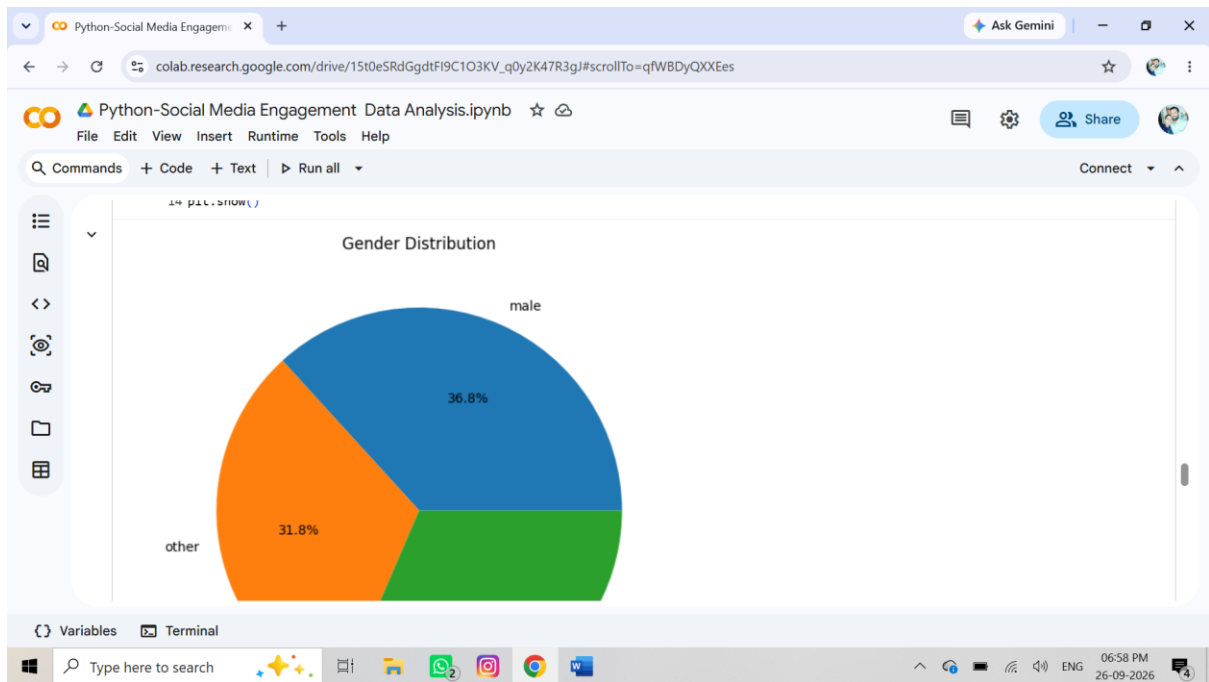
○ Line: daily engagement trend



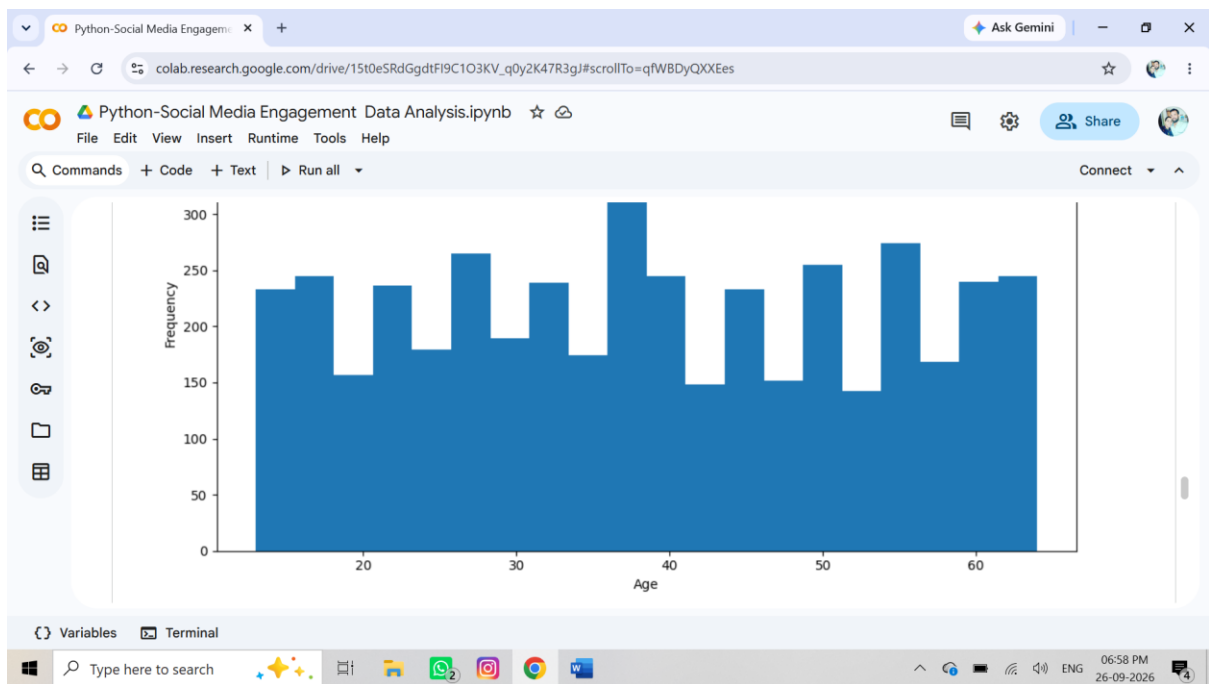
○ Bar: posts by category



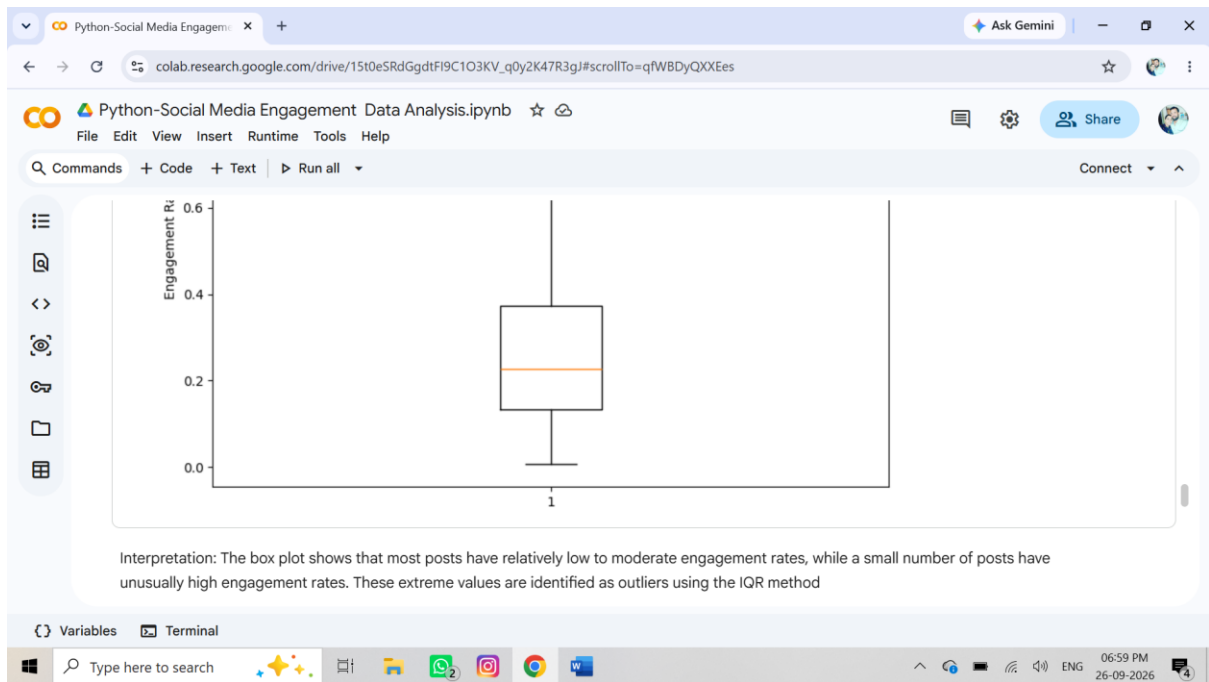
○ Pie: gender distribution



○ **Histogram: age**

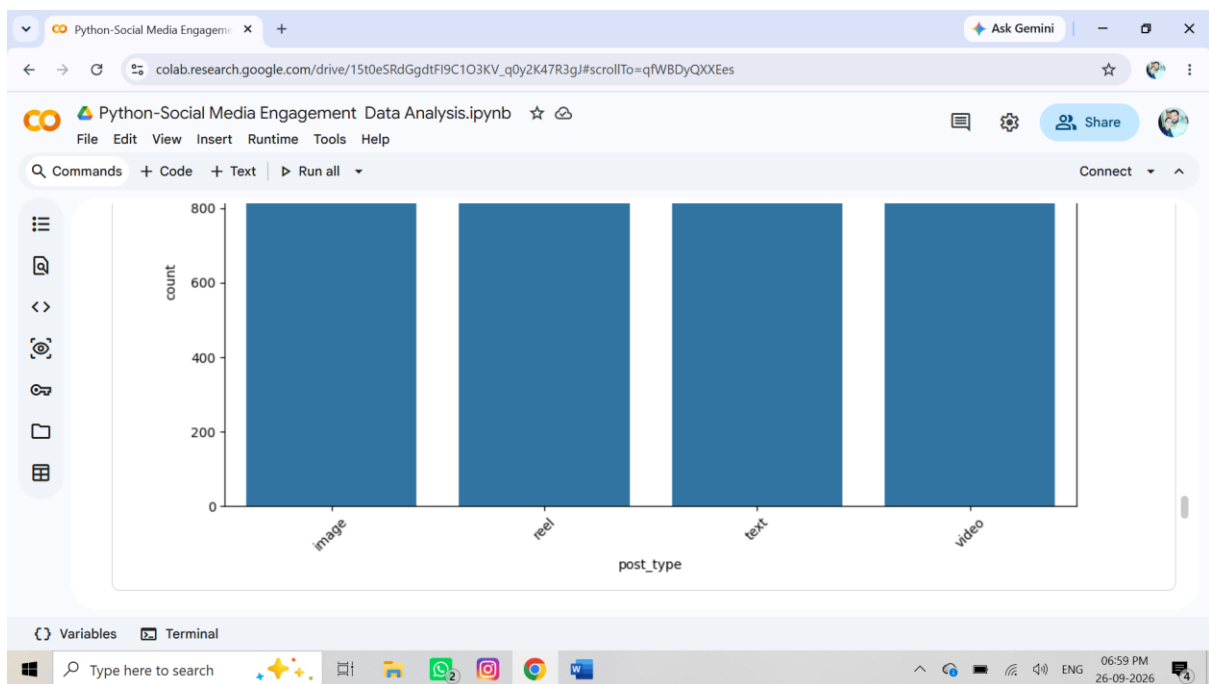


○ **Box: engagement rate**

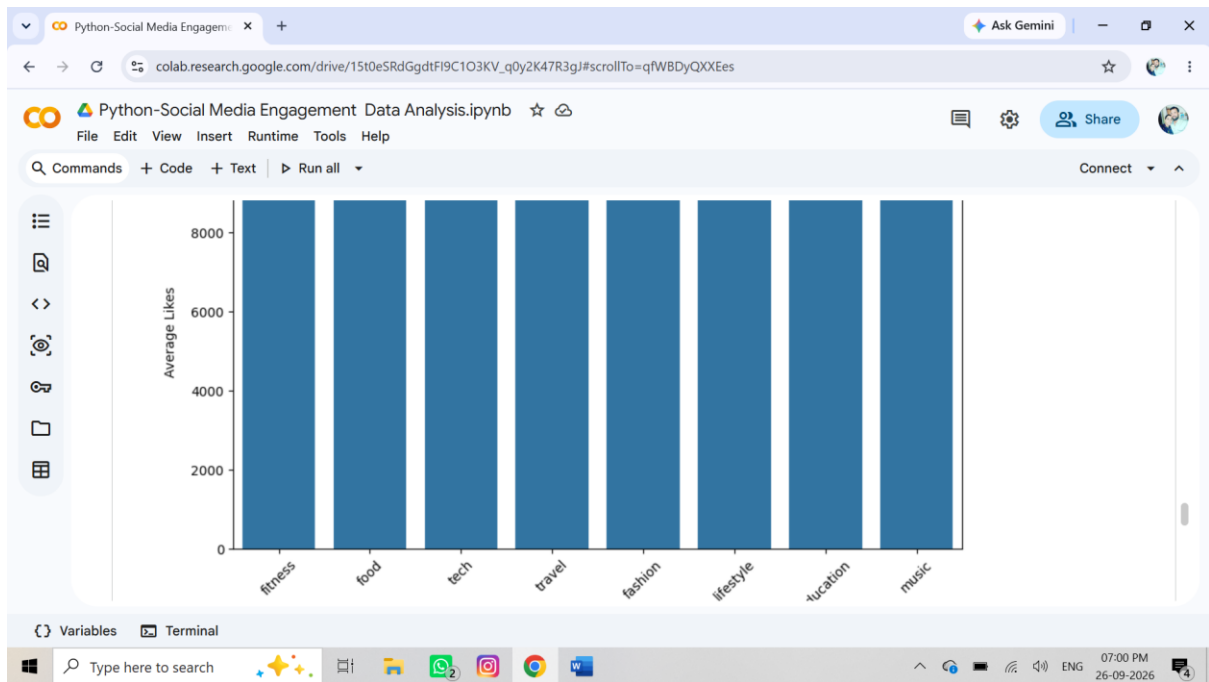


- **Seaborn**

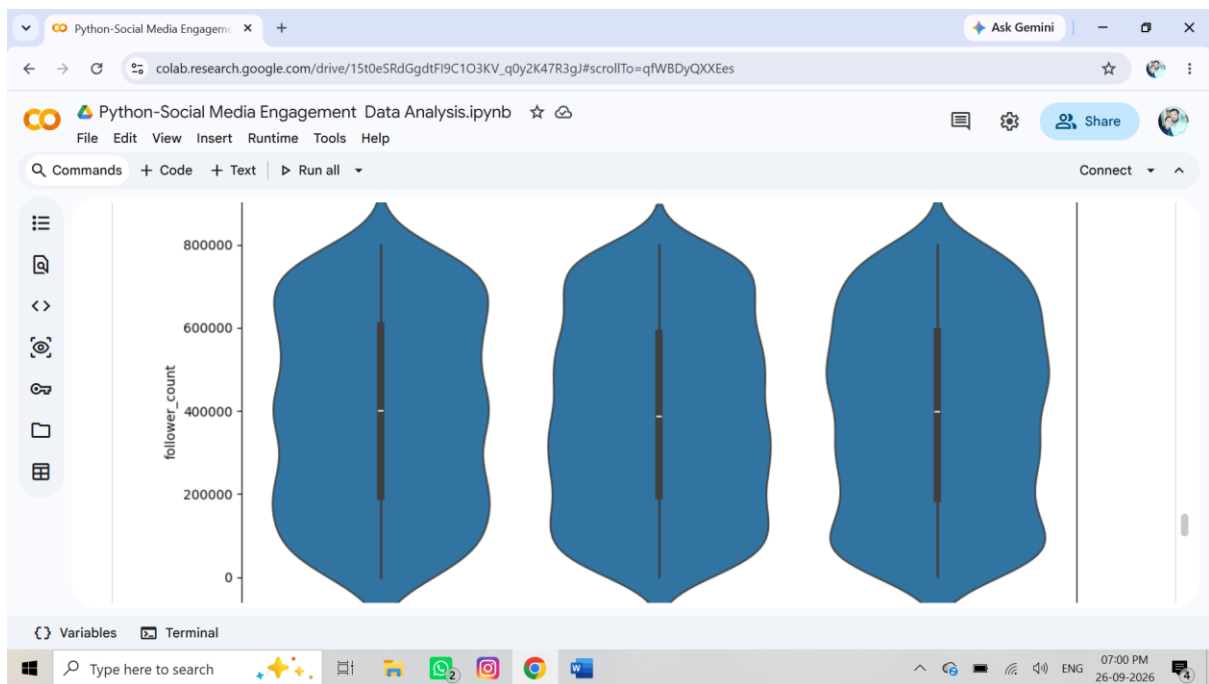
- **Count plot: post type**



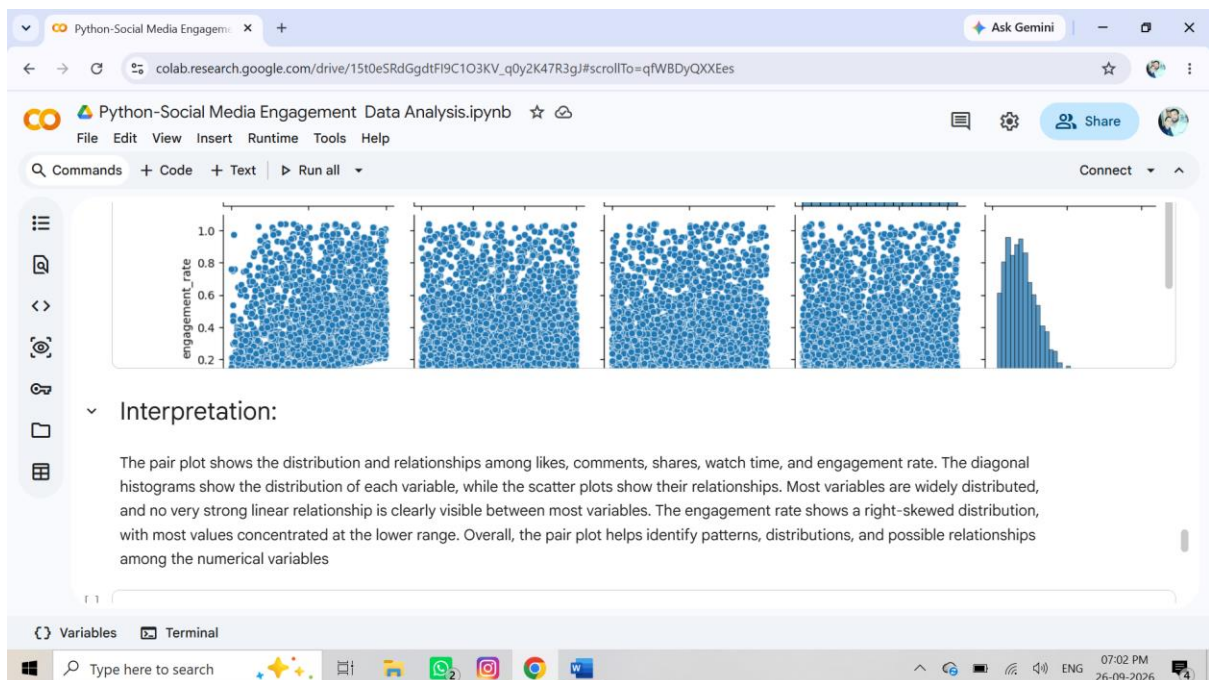
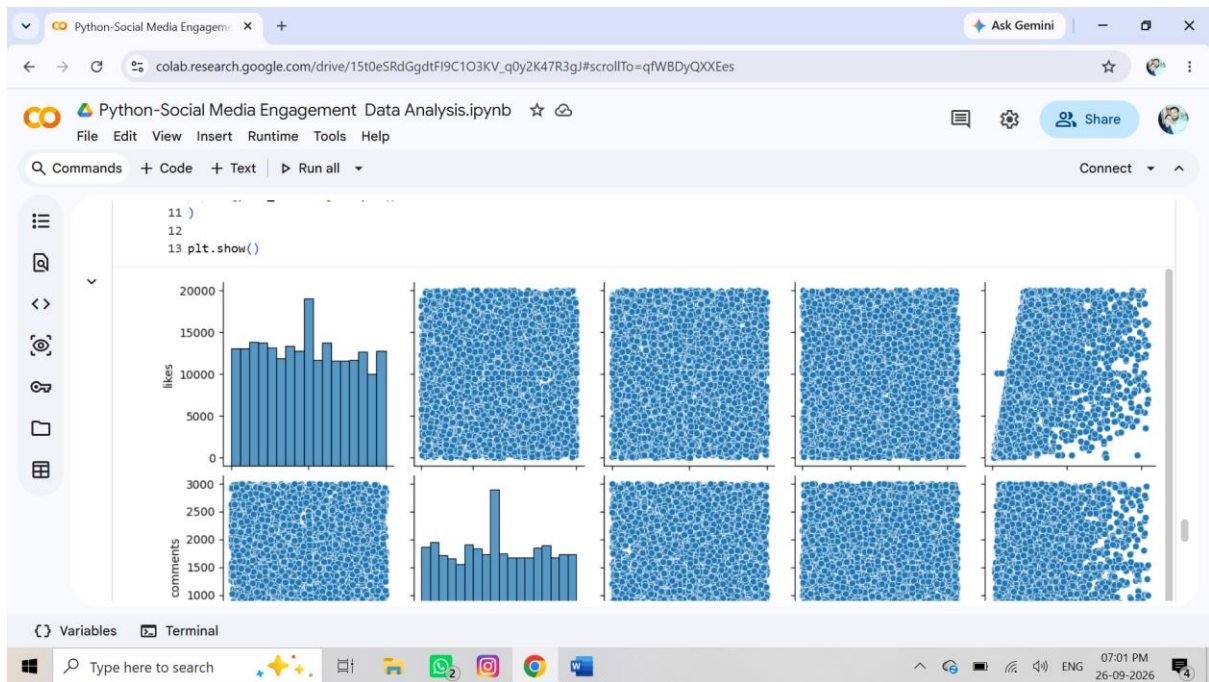
- **Bar plot: avg likes by category**



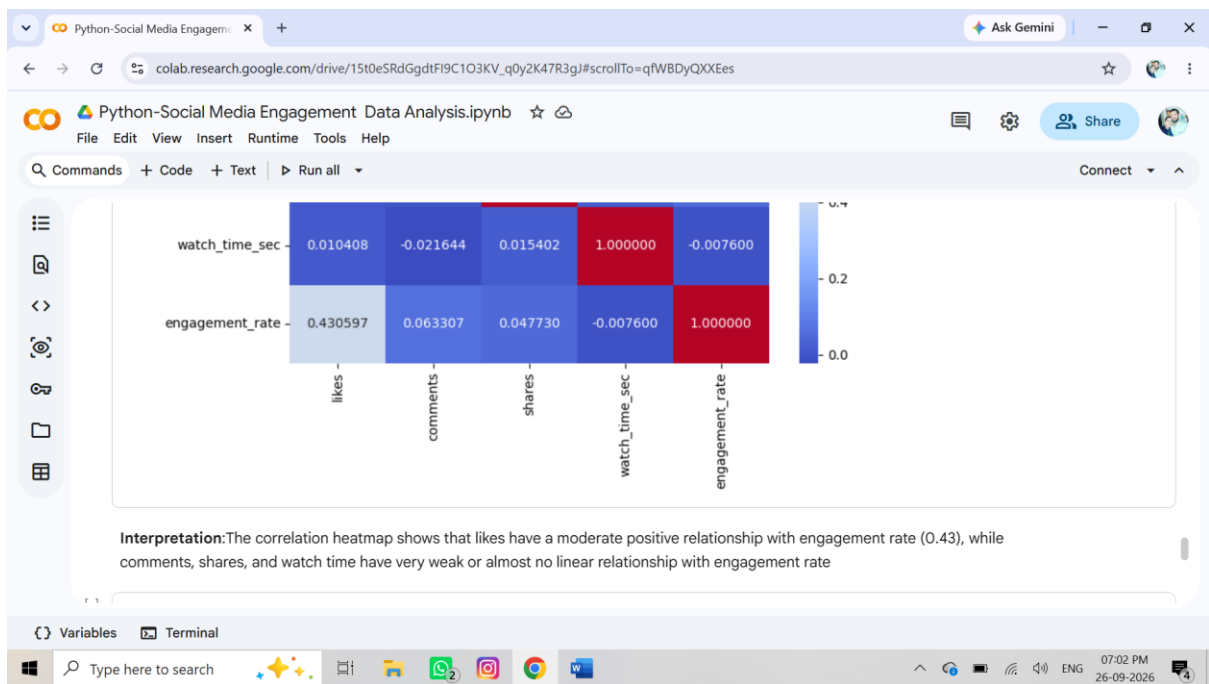
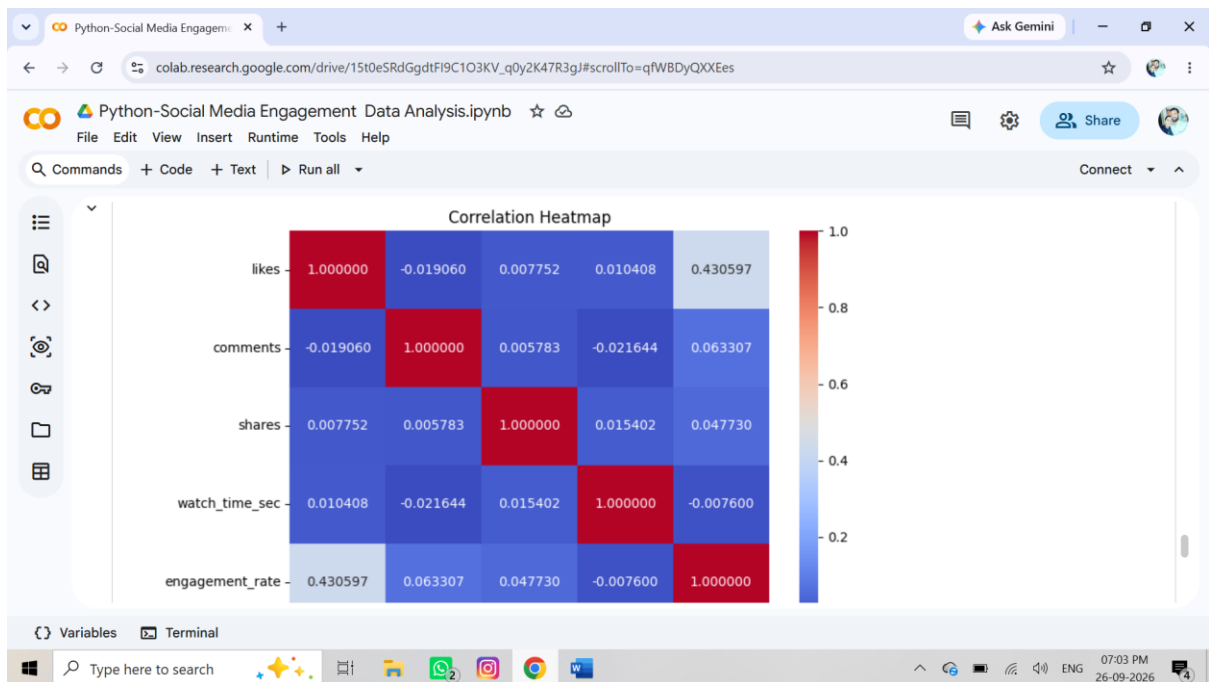
○ Violin: followers vs sentiment



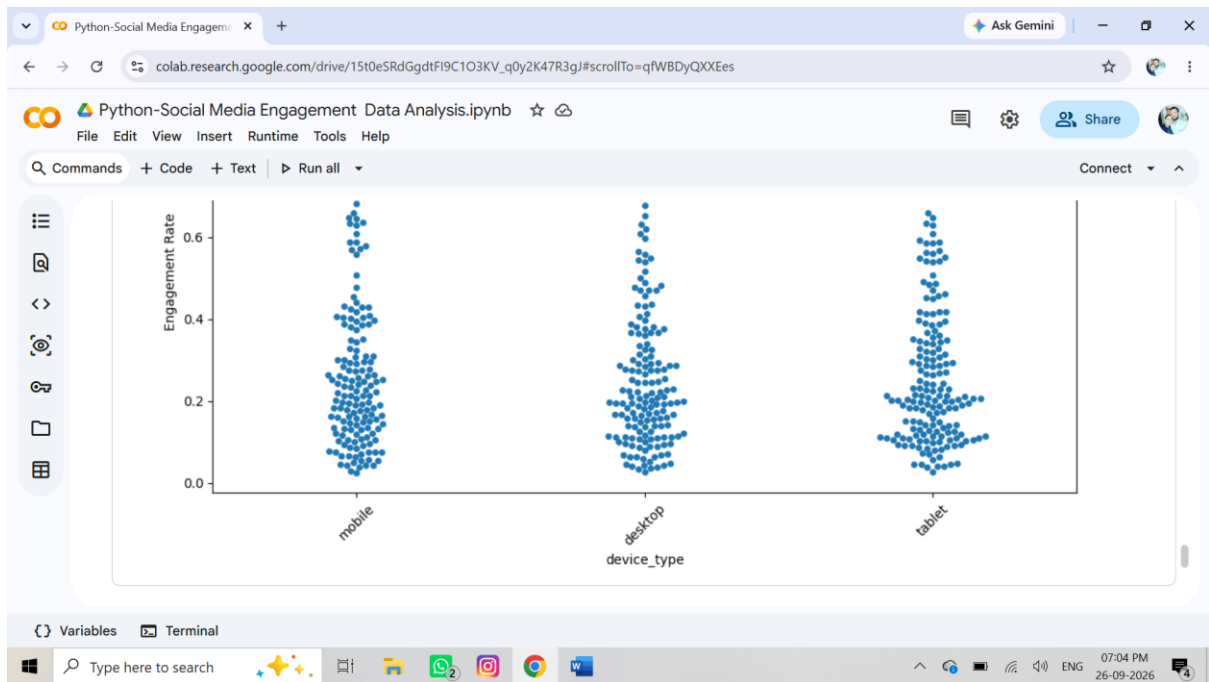
○ Pair plot: numeric features



○ Heatmap: correlation matrix

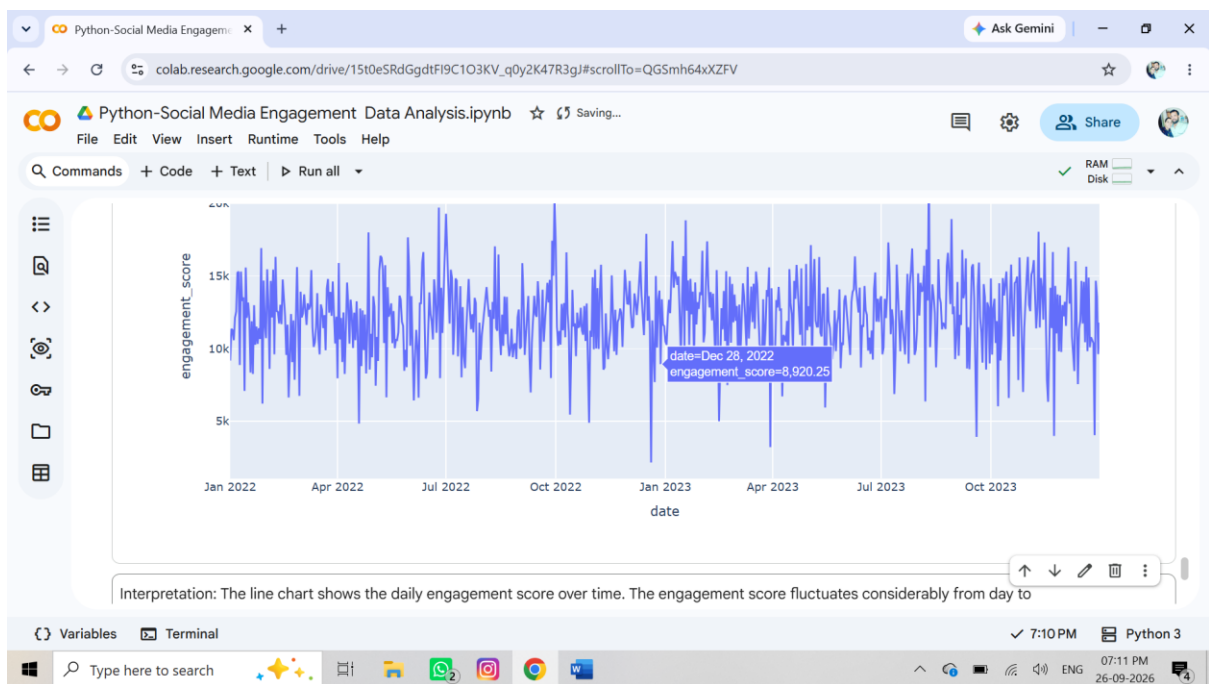


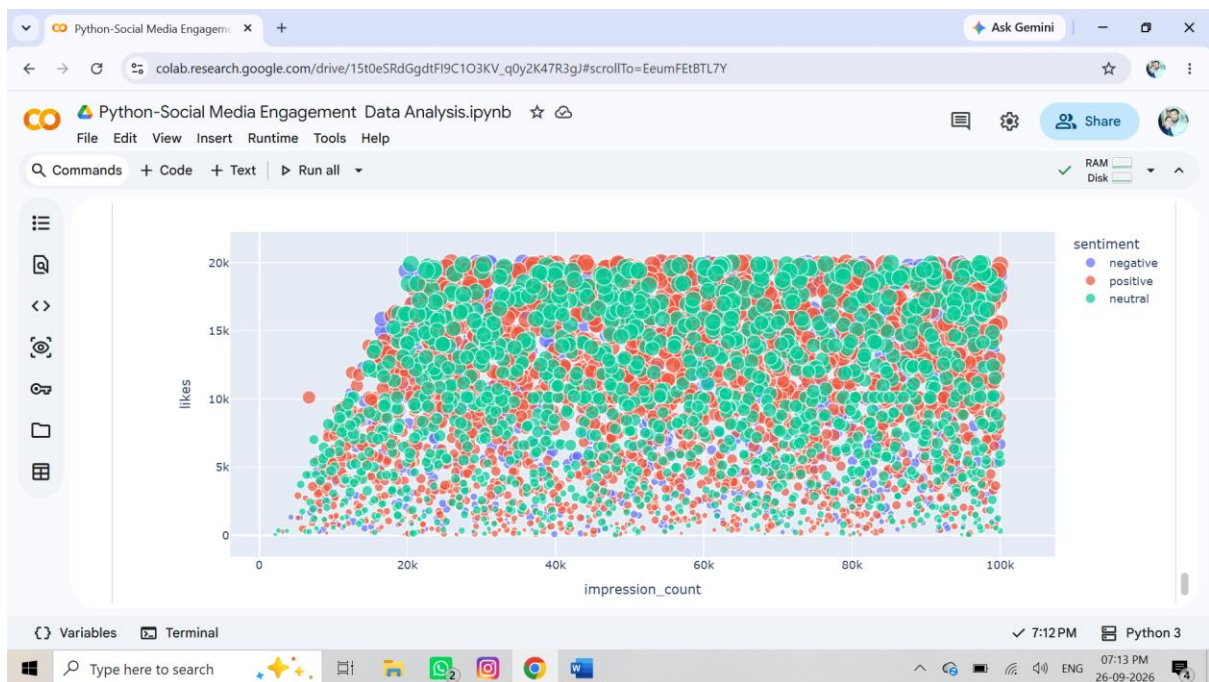
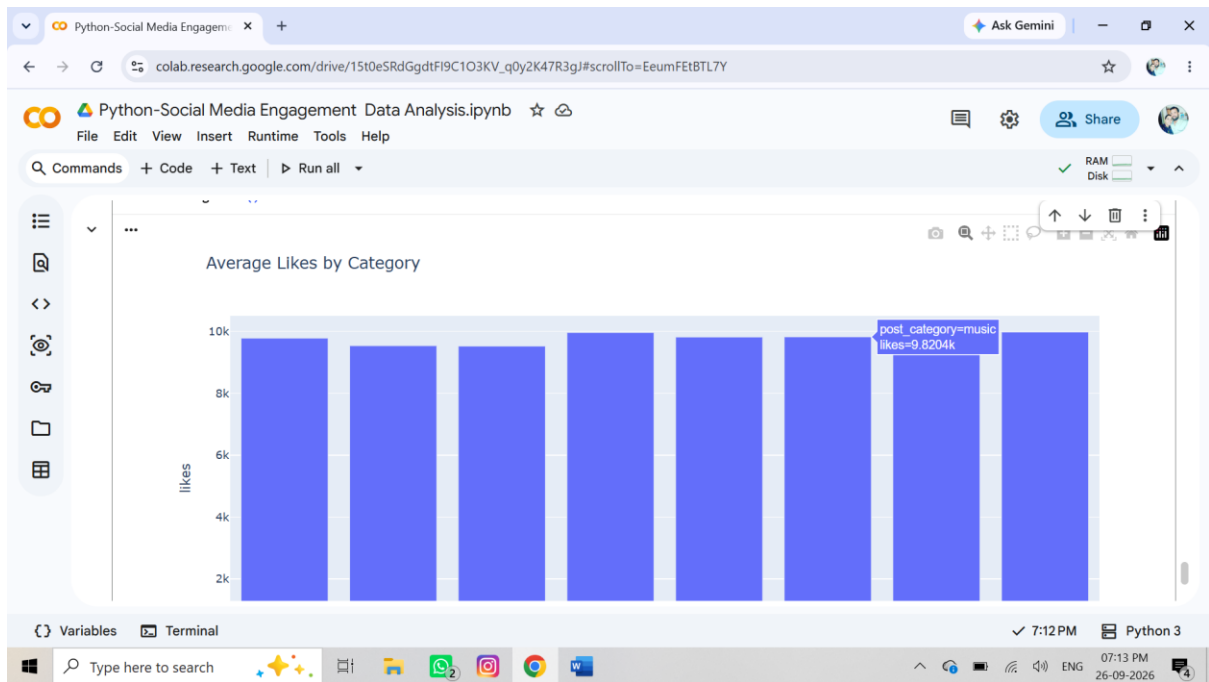
○ Swarm plot: engagement vs device



● Plotly (Interactive)

○ Interactive line chart/bar chart/bubble/scatter chart





Final Insights should include the following analysis:

- **Content Performance**
 - Which post types have the highest engagement?

Python-Social Media Engagement Data Analysis.ipynb

```
1 #Which post types have the highest engagement?
2 post_type_result = df.groupby(
3     "post_type"
4 )["engagement_rate"].mean().sort_values(
5     ascending=False
6 )
7
8 post_type_result
```

post_type	engagement_rate
image	0.293954
text	0.289549
reel	0.286869
video	0.286758

dtype: float64

○ **Best-performing content category?**

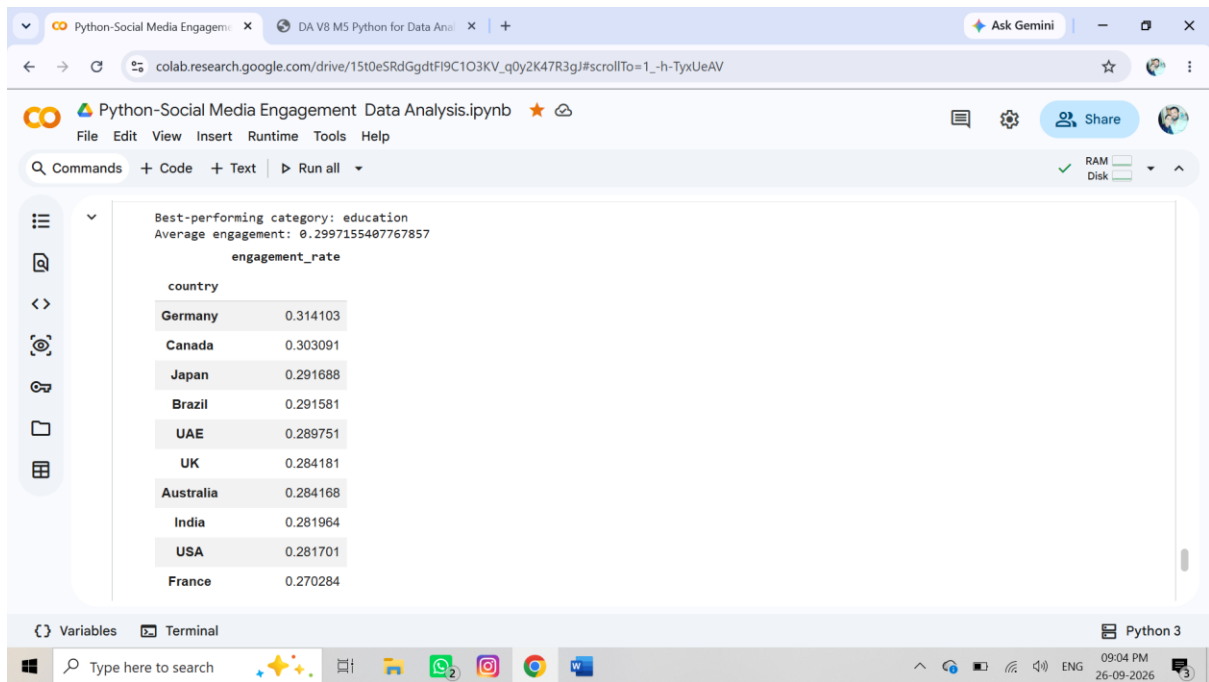
Python-Social Media Engagement Data Analysis.ipynb

```
1 #Best-performing content category?
2 post_category_result = df.groupby(
3     "post_category"
4 )["engagement_rate"].mean().sort_values(
5     ascending=False
6 )
7
8 print(
9     "Best-performing category:",
10    post_category_result.idxmax()
11 )
12
13 print(
14     "Average engagement:",
15    post_category_result.max()
16 )
```

Best-performing category: education
Average engagement: 0.2997155407767857

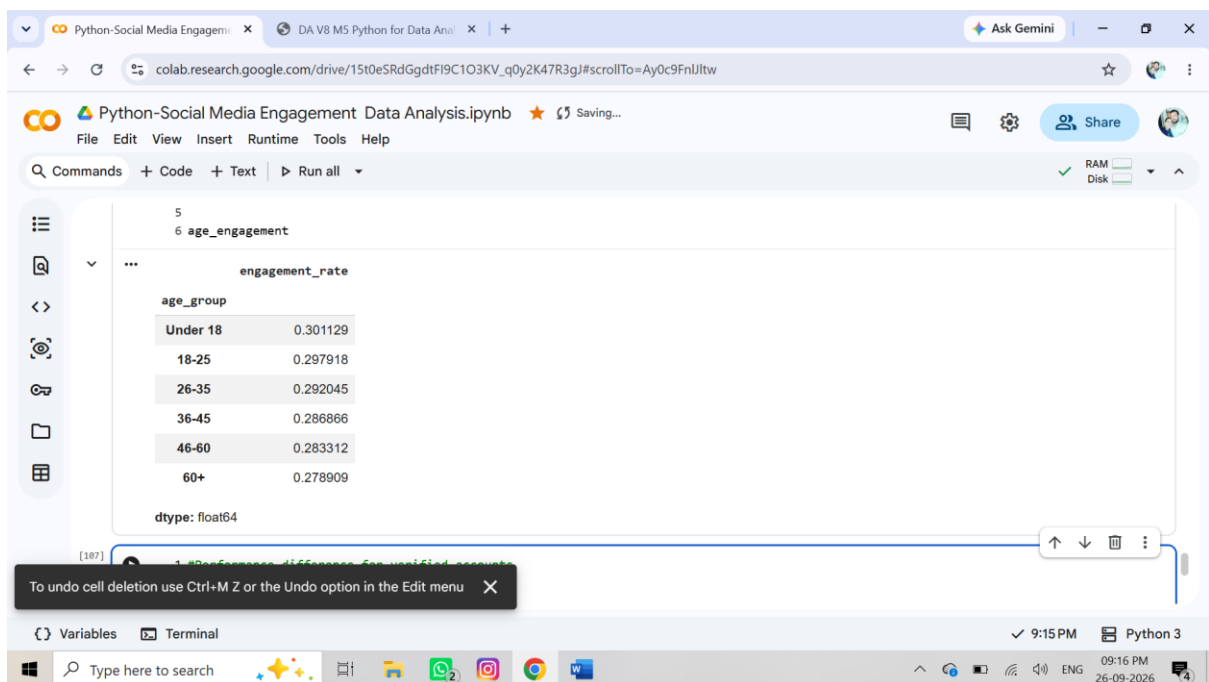
```
1 #Which countries have the highest average engagement rate?
2
```

○ **Which countries have the highest average engagement rate?**



• User Trends

○ How age affects engagement



○ Performance difference for verified accounts

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code in cell [107] calculates the mean engagement rate for verified and unverified accounts. The output is a table showing that verified accounts have a slightly higher mean engagement rate.

```

1 #Performance difference for verified accounts
2 is_verified_result = df.groupby(
3     ["is_verified"]
4 )["engagement_rate"].mean()
5
6 is_verified_result

```

is_verified	engagement_rate
False	0.292488
True	0.259001

dtype: float64

Cell [111] contains the following code:

```

1 #Behavioral Insights
2 #Best time of day for impression_count
3

```

• Behavioral Insights

○ Best time of day for impressions

The screenshot shows the same Google Colab notebook. Cell [111] contains code to analyze the best time of day for impressions based on the 'posted_at' timestamp. The output shows the best hour is 0 (midnight) and the average impression count is approximately 55,581.

```

1 #Behavioral Insights
2 #Best time of day for impression_count
3
4 df["posted_at"] = pd.to_datetime(df["posted_at"], errors="coerce")
5
6 df["hour"] = df["posted_at"].dt.hour
7
8 hourly_impression_count = df.groupby("hour")["impression_count"].mean().sort_values(ascending=False)
9
10 print(hourly_impression_count)
11
12 print("Best hour:", hourly_impression_count.idxmax())
13 print("Average impression count:", hourly_impression_count.max())

```

```

hour
0    55581.162822
Name: impression_count, dtype: float64
Best hour: 0
Average impression count: 55581.1628225237

```

Cell [119] contains the following code:

```

1 #Device device_type impact on watch_time_sec
2

```

○ Device type impact on watch time

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell [122] contains the following Python code:

```
1 device_watch = df.groupby(  
2     "device_type"  
3 )["watch_time_sec"].mean().sort_values(ascending=False)  
4  
5 plt.figure(figsize=(10, 6))  
6
```

The output of the code is a table showing the mean watch time for each device type:

device_type	watch_time_sec
mobile	4077.102961
tablet	3972.656379
desktop	3929.182003

The output also indicates the data type: `dtype: float64`. A notification at the bottom states: "Automatic saving has been pending for 4 minutes. Reloading may fix the problem." The system tray at the bottom shows the time as 9:15 PM on 26-09-2026.

● Sentiment Analysis

○ Which sentiment performs best

The screenshot shows a Google Colab notebook titled "Python-Social Media Engagement Data Analysis.ipynb". The code cell [123] contains the following Python code:

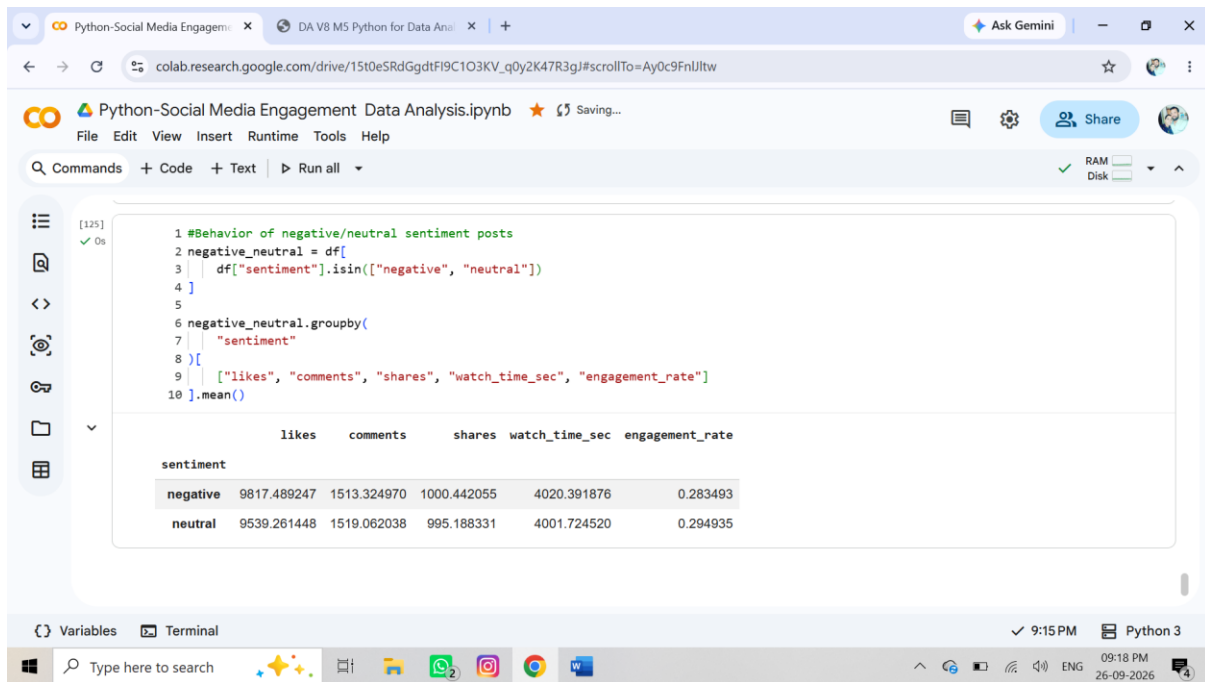
```
1 #Sentiment Analysis  
2 #Which sentiment performs best  
3 sentiment_result = df.groupby(  
4     "sentiment"  
5 )["engagement_rate"].mean().sort_values(  
6     ascending=False  
7 )  
8  
9 sentiment_result
```

The output of the code is a table showing the mean engagement rate for each sentiment:

sentiment	engagement_rate
neutral	0.294935
positive	0.288012
negative	0.283493

The output also indicates the data type: `dtype: float64`. The system tray at the bottom shows the time as 9:18 PM on 26-09-2026.

○ Behavior of negative/neutral sentiment posts



Analysis

1. Descriptive Analysis

Descriptive Analysis is used to summarize and describe the main characteristics of the Social Media Engagement dataset. It provides a basic understanding of the data before performing detailed analysis. In this project, important variables such as likes, comments, shares, watch time, followers, impressions, and engagement rate are summarized using statistical measures. Mean is used to identify the average value, while median represents the middle value of the data. Mode identifies the most frequently occurring value. Percentiles are used to understand how the observations are distributed across different levels. Standard deviation and variance can also be used to understand the spread of the data. Therefore, descriptive analysis provides an overall picture of the dataset and helps understand the general behavior of social media engagement.

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is used to explore the dataset and discover important patterns, relationships, trends, and unusual observations. In this project, the structure of the dataset is examined using functions such as head, tail, shape, columns, info, and dtypes. Categorical variables such as post type, category, country, gender, sentiment, and device are analyzed using unique, nunique, and value_counts. GroupBy analysis is used to compare engagement across different post types, categories, countries, and sentiments. Correlation analysis is also performed to understand the relationships between numerical variables. Outliers and unusual observations are identified during the exploration process. Overall, EDA helps us understand the data more deeply and identify useful patterns that may not be visible from summary statistics alone.

3. Statistical Analysis

Statistical Analysis is used to measure, compare, and interpret the numerical characteristics of the dataset. In this project, statistical analysis is performed on important variables such as likes, comments, shares, watch time, engagement rate, and followers. Measures such as mean, median, and mode are used to understand the central tendency of the data. Standard deviation and variance are used to measure the amount of variation in the observations. Percentiles are used to understand the distribution of values. Skewness is used to determine whether a variable is symmetrically distributed or has a longer tail on one side. Kurtosis helps identify the presence of extreme values and the shape of the distribution. Therefore, statistical analysis provides numerical evidence for understanding the behavior and distribution of social media engagement metrics.

4. Visual and Interactive Analysis

Visual and Interactive Analysis is used to represent the data through charts and graphs so that trends, comparisons, distributions, and relationships can be understood easily. In this project, Matplotlib, Seaborn, and Plotly are used to create different visualizations. Scatter plots are used to study relationships such as likes and impressions. Line charts are used to understand daily engagement trends. Bar charts are used to compare categories and post types. Histograms are used to study age distribution, while box plots are used to identify the distribution and possible outliers in engagement rate. Seaborn is used for count plots, violin plots, pair plots, heatmaps, and swarm plots. Plotly is used to create interactive charts that allow users to explore the data more easily. Overall, visualization makes the analytical results easier to understand and communicate.

Overall Analysis

These four types of analysis work together to provide a complete understanding of the Social Media Engagement dataset. Descriptive Analysis summarizes the data, Exploratory Analysis discovers patterns and relationships, Statistical Analysis measures numerical behavior, and Visual and Interactive Analysis presents the findings in an easy-to-understand graphical form.

Conclusion:

The Social Media Engagement Analysis project demonstrates how raw social media data can be transformed into meaningful information using data analysis techniques.

Through Descriptive, Exploratory, Statistical, and Visual/Interactive Analysis, the project examines content performance, user trends, behavioral patterns, device usage, posting time, and sentiment.

The analysis helps understand how likes, comments, shares, impressions, watch time, followers, and engagement rate vary across different categories of social media activity.

this project shows the importance of data cleaning, statistical analysis, visualization, and interpretation in understanding social media engagement and extracting useful insights from a large dataset